Lab sheet 1

Note: You are advised to use only basic functionalities of JAVA (not build up libraries)

1. You are required to maintain your own record that includes your name, father's name, date of birth, age (computed with respect to the current date). The age includes years, months and days. You are required to show your record on the display screen such as: for displaying your name, it should display like "Name of the student:" followed by its value. Display of each entity should starts with new line. Use scanner class for getting the input from keyboard.
2. You are required to repeat the problem 1 with constraint that, use buffer class (bufferedReader) in place of Scanner class for taking the input from keyboard.
3. You are required to compute perimeter of a rectangle where length and breadth are taken as float and double respectively from the keyboard. Consider length and breadth as integer while computing its perimeter and computed perimeter is displayed as double.
4. You are required to get five integer numbers from the keyboard and check whether these numbers are primes or not.
5. Repeat problem 4 for computing greatest common divisor (GCD) and least common multiple (LCM) for five integers entered from keyboard. (You should not use the concept of array)
6. You are required to convert your marks evaluated out of 100 to the corresponding grades used in MNNIT Allahabad with the use of if-else control statement.
7. Repeat the problem 6 with the use of switch control statement.
8. You are required to model simple digital lock with integer digit combination of three security keys. The combination is set into unlock when it is created security keys. To open the lock user must give correct combination given in term of single digit at a time. The lock explicitly remembers last three digits entered.
9. You are required to model simple digital lock with integer digit combination of three security keys. The combination is set into unlock when it is created security keys. To open the lock user must give correct combination given in term of single digit at a time. The lock instead of remembering last three digits entered to make the lock open. The lock remembers if last digit entered was first security key and if last two digits were first two security keys.